

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and optimized deep learning models for neural data analysis using **PyTorch** and **Python**, achieving an **~0.88 AUC** on held-out validation sets for brain decoding tasks.
- Engineered scalable machine learning pipelines on **AWS** using **Python** for data preprocessing and feature extraction from large, unstructured neural datasets, processing approximately **1,500 events/sec**.
- Collaborated with research scientists to advance **generative modeling** initiatives, contributing to experimentation and hypothesis testing for novel AI applications in neuroscience.
- Improved the reliability of ML inference services by implementing **FastAPI** endpoints with robust input validation and basic feature drift logging, reducing p95 latency to **180ms**.
- Contributed to technical documentation and research activities, including drafting sections of a report on **self-supervised learning** techniques for a distributed team.
- Managed CI/CD processes with **Jenkins** and **Git** for ML model deployment, ensuring code quality and reproducibility across multiple development environments.

SKILLS

ML Frameworks – PyTorch, Hugging Face Transformers, Diffusers, Accelerator, scikit-learn

Programming Languages – Python, SQL, Java, R

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, GPU Systems

Data Handling – Large Datasets, Unstructured Data, Data Warehousing, ETL, Power Query

ML Concepts – Deep Learning, Generative AI, Supervised Learning, Self-Supervised Learning, Representation Learning, Neural Data (EEG/fMRI/MEG concepts)

Engineering Practices – CI/CD (Jenkins), Git, REST APIs, Python Scripting, Automation, Code Quality, Reproducibility

Analytics & Visualization – Power BI, Tableau, DAX, R/RStudio, pandas, NumPy

PROJECTS

Neural Data Decoding Engine (MSc Capstone)

- Developed a deep learning model using **PyTorch** and **Hugging Face Transformers** to decode neural signals from simulated EEG data, achieving **~0.85 AUC** on a 120k-sample dataset.
- Engineered a **Python** data pipeline to ingest, clean, and preprocess raw neural time-series data, handling approximately **800 data points/sec** during simulation.
- Documented the model architecture, training process, and evaluation metrics, contributing to the MSc thesis on applying generative AI to neuroscience.

Automated ML Model Evaluation Framework (HCL Technologies)

- Built a **Python** framework to automate the evaluation of ML models, integrating with **AWS S3** for artifact storage and generating standardized reports.
- Implemented standardized metrics calculation including **AUC** and **F1-score** for supervised classification tasks, reducing manual evaluation time by **40%** across the team.
- Developed **REST APIs** using **Flask** for programmatic submission of model artifacts and retrieval of evaluation results, streamlining the MLOps workflow.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)